Parameter concavity of the generalized inverse hyperbolic sine

Research record: MathDB 325512

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Task 1. Resolve the parameter-concavity assertion

For $p > 0$ and $x > 0$, put

$$F_x(p) = \operatorname{arcsinh}_p(x) = \int_0^x (1 + t^p)^{-1/p}\, dt.$$

The MathDB assertion is that F_x is strictly concave on $(0, \infty)$ for each $0 < x < 1$. This is also the domain in the published Conjecture 2.4 of Baricz, Bhayo and Vuorinen [1]. A later survey [2], Conjecture 5.1, states the narrower domain $(1, \infty)$. We distinguish these two versions explicitly: the published global assertion is false, and the narrower assertion is true.

Theorem 1 (An exact midpoint counterexample)**.** *At $x = 1/4$, $p_- = 1/32$, $p_0 = 1/16$, and $p_+ = 3/32$,*

$$F_{1/4}(p_0) < \frac{1}{65536} < \frac{1}{16384} < \frac{F_{1/4}(p_-) + F_{1/4}(p_+)}{2}.$$

Since $p_0 = (p_- + p_+)/2$, this contradicts concavity.

Proof. The arithmetic–geometric mean inequality gives, for $t > 0$,

$$1 + t^p \geq 2t^{p/2}, \qquad (1 + t^p)^{-1/p} \leq 2^{-1/p} t^{-1/2}.$$

Equality requires $t = 1$, so integration gives

$$0 < F_x(p) < 2\sqrt{x}\, 2^{-1/p}. \tag{1}$$

For $x = 1/4$ and $p = p_0$ this is $F_{1/4}(p_0) < 2^{-16}$. On $0 < t < 1/4$, we also have $1 + t^{3/32} < 2$. Consequently

$$(1 + t^{3/32})^{-32/3} > 2^{-32/3} > 2^{-11},$$

and therefore $F_{1/4}(p_+) > 2^{-13}$. Since $F_{1/4}(p_-) > 0$, the midpoint average exceeds 2^{-14}. All bounds are strict and use only the stated rational parameters. □

Proposition 2 (Failure for every fixed argument)**.** *For every $x > 0$, F_x cannot be concave on any interval $(0, b)$ with $b > 0$.*

Proof. By (1), $F_x(p) \to 0$ and $F_x(p)/p \to 0$ as $p \downarrow 0$. If F_x were concave on $(0, b)$, then for $0 < r < s < b$, taking the limit of the concavity inequality with one endpoint tending to zero would give $F_x(r) \geq (r/s)F_x(s)$. Thus $F_x(s)/s \leq \lim_{r \downarrow 0} F_x(r)/r = 0$, contradicting positivity. □

Theorem 3 (A positive range, including the survey conjecture)**.** *For each $x > 0$, F_x is strictly increasing on $(0, \infty)$ and strictly concave on $[\frac{1}{2}\log 2, \infty)$. In particular, the conjecture restricted to $p > 1$ is true.*

Proof. For fixed $t > 0$ set $h(p, t) = (1 + t^p)^{-1/p}$, $s = -p \log t$, and

$$A(s) = \log(1 + e^{-s}) + \frac{s}{1 + e^s}, \qquad B(s) = \frac{e^s}{(1 + e^s)^2}.$$

Direct differentiation gives

$$A'(s) = -sB(s), \qquad A(-s) = A(s), \qquad 0 < A(s) \le A(0) = \log 2.$$

The positivity and the limit $A(s) \to 0$ as $|s| \to \infty$ also follow directly from the definition. Since $\partial_p \log h = A(s)/p^2$, we have $\partial_p h > 0$ and

$$\frac{\partial_p^2 h}{h} = \frac{A(s)^2 - p\{2A(s) + s^2 B(s)\}}{p^4}. \tag{2}$$

If $p \ge \frac{1}{2}\log 2$, then

$$A(s)^2 - 2pA(s) \le A(s)\{A(s) - \log 2\} \le 0.$$

Together with the nonpositive remaining term $-ps^2 B(s)$, this proves $\partial_p^2 h \le 0$. Equality can occur only when $p = \frac{1}{2}\log 2$ and $t = 1$. Integration over any interval $(0, x)$ therefore gives $F_x''(p) < 0$ throughout the stated range.

For completeness, differentiation under the integral is justified on every compact parameter interval in $(0, \infty)$: $0 < h \le 1$, A is bounded, and $s^2 B(s)$ is bounded on the real line. The displayed first and second derivatives consequently have a uniform integrable constant bound on the finite interval $(0, x)$. □

Scope. The counterexample settles the global ordinary-concavity question actually listed in MathDB and the published paper. Theorem 3 also proves the narrower conjecture retained in the later survey, so the result does not depend solely on a discrepancy in the lower parameter endpoint. The survey's consequent Turán inequality for $p > 2$ follows from strict concavity and the arithmetic–geometric mean inequality. The separate published Turán assertion for every $p > 1$ is not refuted by the ordinary-concavity counterexample and is not needed for the MathDB question.

Eligibility. The accompanying literature audit records fresh open/empty-claim MathDB checks, the published source, the later survey, relevant citing papers and public claim catalogues. No matching full public claim was located in that bounded search.

Task 2. Determine the sharp concavity threshold

For fixed $x > 0$, determine the least p_x such that F_x is concave on $[p_x, \infty)$. Find the best threshold uniform for $0 < x < 1$.

Task 3. Locate and count the inflection points

Determine whether F_x'' has exactly one zero on $(0, \infty)$ for each $x > 0$, and describe how such zeros depend on x.

Task 4. Obtain uniform small-parameter asymptotics

Derive a full asymptotic expansion of $2^{1/p}F_x(p)$ as $p \downarrow 0$, with explicit remainder bounds uniform on specified ranges of x, including regimes in which x also tends to zero.

Task 5. Analyze affine parameter paths

For $\operatorname{arcsinh}_{p,q}(x) = \int_0^x (1+t^q)^{-1/p}\,dt$, classify ordinary concavity along $q = \alpha p + \beta$ on intervals where $p, q > 0$, and identify sharp conditions on (α, β) and x.

References

[1] Á. Baricz, B. A. Bhayo and M. Vuorinen, *Turán type inequalities for generalized inverse trigonometric functions*, Filomat **29**(2) (2015), 303–313. doi:10.2298/FIL1502303B. Conjecture 2.4, p. 306.

[2] L. Yin, L.-G. Huang, Y.-L. Wang and X.-L. Lin, *A survey for generalized trigonometric and hyperbolic functions*, Journal of Mathematical Inequalities **13**(3) (2019), 833–854. doi:10.7153/jmi-2019-13-58. Conjecture 5.1.